

Agri-DINO: Domain-Adaptive Self-Supervision for Fine-Grained Plant Recognition

Md Min-Ha-Zul Abedin Md Mehedi Hasan
Department of Computer Science and Software Engineering
Auburn University
{mza0288, mzh0159}@auburn.edu

Abstract

*General-purpose vision foundation models are strong at broad recognition, but agricultural images often require decisions from subtle, local cues such as leaf venation, small lesions, and phenotypic variation across environments. We introduce **Agri-DINO**, a DINOv3-initialized Vision Transformer adapted to agriculture through domain-adaptive pre-training (DAPT) on an assembled corpus of 5.17 million botanical and phytopathological images. The additional agricultural pre-training uses no downstream human labels. For downstream transfer, we evaluate a structured protocol that combines **two-stage linear probing then fine-tuning (LP-FT)**, **layer-wise learning-rate decay (LLRD)**, ($\gamma=0.75$), and **256 $\times$ 256 tokenization**. The protocol is designed to limit feature drift on small downstream datasets. On a fixed stratified split of a fine-grained medicinal-plant classification dataset, Agri-DINO reaches **59.67% top-1 accuracy**, a **+10.32-point** gain over random initialization and similar accuracy to supervised ImageNet-21k initialization. The cumulative ablation quantifies the incremental contribution of each component. We also diagnose a benchmark-construction error that fragments disease labels into sample-specific micro-classes and makes the resulting classification problem appear substantially harder. The study is a controlled transfer analysis; it does not establish cross-domain robustness or universal superiority over generic pre-training.*

1. Introduction

Plant diseases and pests destroy an estimated \$220 billion of the global agricultural harvest annually [1], while access to expert plant pathologists remains limited. Visual diagnosis from smartphones, field cameras, and drones could help, but only if models can resolve fine-grained signals such as early chlorosis, marginal necrosis, and spore-colony morphology under changing illumination, viewpoint, and background

conditions.

The dominant transfer-learning recipe in agricultural vision is to initialize a convolutional or transformer backbone on ImageNet and fine-tune it on a small, task-specific plant dataset. This recipe is useful, but it leaves two gaps. First, ImageNet’s visual statistics are only an imperfect match for plant tissue and field acquisition. Second, supervised pre-training requires large labeled corpora, whereas botanical and phytopathological annotation requires specialized expertise [2]. Self-supervision removes the label requirement for pre-training, but it does not remove the adaptation problem: small downstream datasets can induce feature drift, and the standard 224×224 input can undersample fine biological structure.

This work studies these issues through the following components of Agri-DINO:

C1. DINOv3-Initialized Agricultural Adaptation. Starting from an official DINOv3 ViT-Base/16 checkpoint, we stream 5.17 million images spanning crop diseases, weed species, remote-sensing fields, and botanical specimens through a DINO-style student–teacher distillation pipeline. The resulting 86M-parameter ViT-Base/16 checkpoint is adapted to agricultural visual statistics without using the source labels.

C2. Two-Stage LP-FT Adaptation Protocol. We evaluate a two-stage protocol: a brief linear-probing warmup to align the classification head, followed by backbone fine-tuning with layer-wise learning rate decay. In the cumulative ablation, the full protocol is 24.48 points above the no-LP, low-resolution, no-TTA baseline; this sequence is not an independent estimate of forgetting or of any single component’s effect.

C3. Evaluation Protocol and Benchmark Audit. We define standardized train/validation splits and evaluation scripts for downstream benchmarking, and we expose a micro-class fragmentation pathology in one YOLO-to-classification conversion that can distort the apparent task difficulty. We use “Agri-Bench” as shorthand for this protocol and audit, not as a claim of a fully released multi-task benchmark suite.

On the reported medicinal-plant split, Agri-DINO performs similarly to supervised ImageNet-21k initialization under the reported adaptation protocols, without using downstream human labels during additional pre-training. We treat this result as evidence for the value of in-domain self-supervision on this split, not as a claim that one benchmark establishes general agricultural superiority.

2. Related Work

2.1. Agricultural Recognition and Benchmarks

Plant recognition is a fine-grained, context-sensitive problem: class identity may depend on small morphological cues or distributed symptoms, while images vary with crop stage, organ, illumination, viewpoint, and field background [6, 7]. The commonly used benchmarks cover different regimes. PlantVillage is a controlled, single-leaf disease benchmark [23], whereas iNaturalist and PlantCLEF emphasize large-vocabulary species recognition in more natural imagery [24, 25]. Their scores are therefore not interchangeable: the benchmarks differ in capture conditions, taxonomy, label quality, and even the unit of prediction. Dataset reviews also identify inconsistent splits, annotation noise, and leakage through related images as sources of optimistic estimates [8, 9]. We make these protocol choices explicit and treat PlantSeg as a data-quality audit.

2.2. Self-Supervised Pre-Training

Self-supervised learning (SSL) removes the need for human labels during pre-training. Contrastive methods such as SimCLR [20] learn agreement between augmented views while separating other images. Masked image modeling (MIM) methods such as BEiT [18] and MAE [19] instead predict masked tokens or pixels, which is attractive for learning local structure and dense representations. DINO [15] uses a momentum teacher and centered cross-view predictions; its features exhibit semantic neighborhoods and spatial organization without an explicit negative-pair queue. iBOT [17] adds masked-patch prediction, while DINOv2 [16] scales the recipe through careful curation and self-distillation. Our backbone is initialized from the official DINOv3 [5] ViT-Base/16 checkpoint before the additional agricultural adaptation.

We choose DINO as the primary objective because the downstream task is fine-grained classification: semantic separability and local-neighborhood structure are more directly aligned with the target than reconstruction fidelity alone. This is a task-motivated choice, not a claim that DINO dominates every SSL setting. Our corpus is roughly 27 times smaller than LVD-142M, so coverage of agricultural acquisition conditions is an important complement to scale.

2.3. Domain Adaptation

Domain-aligned pre-training has improved transfer in specialized domains such as pathology and remote sensing, where acquisition physics, field of view, and cross-scene variation differ from generic web imagery [12, 13]. Agricultural studies point in the same direction. AgriNet and PDDD-PreTrain show that plant-domain representations can improve disease and pest transfer over generic initialization [10, 11]; unsupervised adaptation and agricultural foundation models further show that unlabeled in-domain images can help bridge laboratory-to-field shifts [3, 4]. Evidence remains fragmented across crops, tasks, and protocols, and few studies isolate self-supervised domain adaptation from downstream fine-tuning on a fixed backbone. Agri-DINO addresses this narrower, controlled question.

2.4. ViT Fine-Tuning

End-to-end fine-tuning of a large ViT on a small dataset can overfit and distort useful pre-trained features [22]. LP-FT first fits the classification head with the backbone frozen, then permits controlled backbone updates; the staged optimization reduces the early feature drift caused by a randomly initialized head. Layer-wise learning-rate decay (LLRD) complements this schedule by assigning smaller updates to earlier blocks. Parameter-efficient alternatives such as BitFit [26] and low-rank updates can reduce memory and storage, and can preserve pre-training-domain performance in some settings [14], but add method-specific design choices. We study LP-FT with LLRD because it preserves full adaptation while remaining simple to inspect in the cumulative ablation; the goal is transfer robustness, not a PEFT comparison.

2.5. Benchmarking and Positioning

Benchmark design is part of the scientific problem. VTAB [27] shows that representation quality depends on the task suite, and agricultural benchmarks add further variation in class granularity, split strategy, preprocessing, and label quality. We therefore report the split, taxonomy, and input protocol explicitly and include a deterministic evaluation harness. Our contribution is a controlled joint study of DINO-based domain-adaptive pre-training, LP-FT/LLRD adaptation, and benchmark construction on a common ViT-Base/16 backbone—not a claim of universal agricultural superiority.

3. Agri-DINO

Figure 1 summarizes the two phases of our framework: (i) domain-adaptive self-supervised pre-training and (ii) structured two-stage fine-tuning.

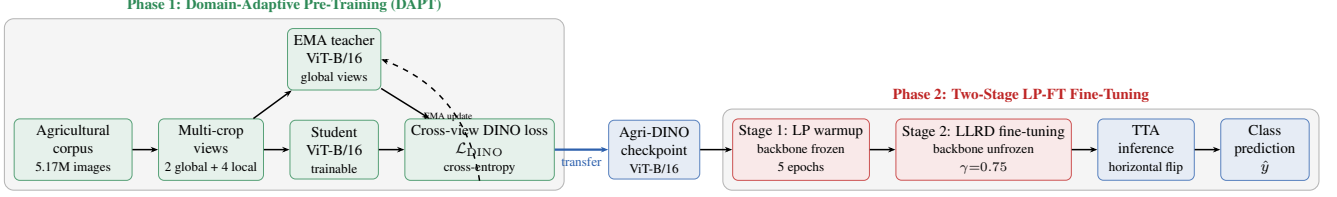


Figure 1. **Agri-DINO pipeline.** *Phase 1 (green):* Two global and four local views of each unlabeled agricultural image train a ViT-Base/16 student with DINO cross-view distillation; the teacher is updated by exponential moving average (EMA). *Phase 2 (red):* The resulting checkpoint is adapted with a five-epoch linear-probing warmup, followed by end-to-end fine-tuning with LLRD ($\gamma=0.75$) and horizontal-flip TTA. Solid arrows show the training path; the dashed arrow denotes the EMA update.

3.1. Pre-Training Corpus Construction

Our corpus contains 5,170,016 RGB images from 38 publicly available agricultural and botanical datasets. It spans (i) foliar disease classification (PlantVillage, PlantDoc, and a crop-disease dataset), (ii) weed and crop-species identification (DeepWeeds and an iNaturalist crop subset), (iii) drone-captured field imagery, and (iv) general botanical photography. We shard the images into `.tar` WebDataset archives so the pre-training loader can stream samples without materializing the full corpus. The current corpus is neither class-balanced nor deduplicated; we leave these operations for a controlled follow-up study and treat the present result as a baseline for raw-corpus domain adaptation.

3.2. Self-Supervised Pre-Training Objective

Let $\mathcal{D} = \{x_i\}_{i=1}^N$ denote the unlabeled pre-training corpus. We instantiate a student network with parameters θ_s and a momentum teacher with parameters θ_t , initialized from the same official DINOv3-pre-trained ViT-Base/16 checkpoint [5]. This initialization makes our method an additional agricultural adaptation stage rather than a from-scratch pre-training run. For each image x , the augmentation pipeline produces two global views $V_g = \{v_1^g, v_2^g\}$ (covering at least 50% of the image area) and four local views V_l (covering 20–50%). The student processes all views, while the teacher processes only the global views. We minimize the cross-entropy between the teacher and student output distributions:

$$\mathcal{L}_{DINO} = \sum_{v \in V_g \cup V_l} \sum_{\substack{v' \in V_g \\ v' \neq v}} H(P_t(v'), P_s(v)), \quad (1)$$

where $H(a, b) = -a \log b$ is the cross-entropy and P_s, P_t are softmax-normalized projection-head outputs. We center the teacher outputs before the softmax, following the DINO training recipe. Teacher parameters are updated with an exponential moving average:

$$\theta_t \leftarrow \lambda \theta_t + (1 - \lambda) \theta_s, \quad \lambda \in [0.996, 1.0], \quad (2)$$

where λ follows a cosine schedule that approaches 1 over training.

3.3. Two-Stage LP-FT with Domain-Calibrated LLRD

Motivation. When a self-supervised ViT is fine-tuned end-to-end on a small downstream dataset $\mathcal{D}_{\text{down}}$, the randomly initialized classification head can produce large early gradients before it has aligned with the frozen feature space. Such updates can distort pre-trained features and reduce out-of-distribution robustness [22]. We therefore separate head alignment from backbone adaptation. In our cumulative ablation, the full protocol is 24.48 points above the no-LP, low-resolution, no-TTA baseline; this difference includes several interventions and is not interpreted as a direct measurement of forgetting.

Stage 1: Linear Probing (Epochs $1 \leq e \leq E_w$). The backbone parameters θ_{bb} are strictly frozen:

$$\nabla_{\theta_{\text{bb}}} \mathcal{L} = \mathbf{0}, \quad e \leq E_w = 5. \quad (3)$$

Only $W_{\text{head}} \sim \mathcal{N}(0, 0.02^2)$ is updated via AdamW. This warmup aligns the classification hyperplane with the frozen DINO feature space before any gradient signal reaches the backbone.

Stage 2: End-to-End LLRD Fine-Tuning ($e > E_w$). The backbone is unfrozen and Layer-wise Learning Rate Decay (LLRD) is applied. For transformer block $l \in \{1, \dots, L\}$ (with $L = 12$ for ViT-Base), the effective learning rate is:

$$\eta_l = \eta_{\text{base}} \cdot \gamma^{L+1-l}, \quad (4)$$

where $\gamma = 0.75$ is the decay factor used in our experiments. This assigns *lower* learning rates to early blocks and *higher* rates to upper blocks. We use this value as a practical setting for the present benchmark; a broader sweep is needed before treating it as a universal calibration for self-supervised features.

3.4. High-Resolution Tokenization and Test-Time Augmentation

Botanical species discrimination can depend on fine structural cues—leaf margin serrations, trichome density, and

vein branching patterns—that are spatially undersampled at the standard 224×224 resolution ($N_p = 196$ patches). We therefore use 256×256 inputs, increasing the sequence length to $N_p = 256$ patches; the ViT positional embeddings are interpolated to this grid.

At inference, we apply **test-time augmentation (TTA)** by averaging the pre-softmax logits for the original image and its horizontal reflection:

$$\hat{y} = \arg \max \left[\frac{1}{2} (f_\theta(x) + f_\theta(\text{flip}(x))) \right]. \quad (5)$$

We treat the horizontal flip as label-preserving for the image-level taxonomy used here. This assumption should be revisited for tasks where orientation itself is diagnostic.

4. Experiments

We organize the evaluation around four reader-facing questions: how Agri-DINO compares with standard initialization, which adaptation components matter, how the training dynamics evolve, and whether the PlantSeg benchmark is well-formed.

4.1. Datasets and Implementation

Downstream datasets. We evaluate on two benchmarks:

- **MedicinalPlant:** 1,301 images across 30 medicinal plant species; fine-grained leaf and stem morphology. 80%/20% train/validation split stratified by class.
- **PlantSeg:** Crop disease images originally distributed as oriented bounding-box (OBB) YOLO annotations, converted to an image-classification layout. We report a fragmented-label audit and analyze a systematic labeling pathology (§4.5); the corrected taxonomy is not yet evaluated.

Implementation details. All experiments run on a single NVIDIA RTX 4090 (24 GB VRAM) under PyTorch 2.0 with `bfloat16` mixed precision. We use the AdamW optimizer with base learning rate $\eta = 3 \times 10^{-4}$, weight decay 0.1, label smoothing 0.2, batch size 128, and 50 training epochs. A cosine annealing schedule governs η after a 5-epoch warmup. Gradient clipping at norm 1.0 and dropout (rate 0.1) are applied throughout. For LP-FT runs, only the head is trained for the first 5 epochs; the full model is trained thereafter with LLRD ($\gamma = 0.75$).

4.2. Comparison with Baseline Initialization Strategies

Table 1 compares Agri-DINO with two initialization baselines on MedicinalPlant. All rows use the same ViT-Base/16 architecture and input resolution; the reported baselines use direct end-to-end fine-tuning, whereas Agri-DINO uses LP-FT.

Key findings. Agri-DINO reaches **59.67%** top-1 accuracy on this split, an absolute improvement of **+10.32 points**

Table 1. **Main results on MedicinalPlant.** Top-1 validation accuracy on the fixed stratified split; all models use ViT-Base/16 at 256×256 . “Adaptation” identifies the reported downstream protocol. Δ is in percentage points. **Bold:** best result; Underlined: second-best. †: no human labels are used during pre-training.

Method	Pre-training	Adaptation	Val. Acc. (%)	Δ vs. scratch (pp)
Scratch (Random)	—	Direct FT	49.35	—
ImageNet-21k (Sup.)	21k labels	Direct FT	<u>59.99</u>	+10.64
Agri-DINO (Ours)†	SSL	LP-FT	59.67	+10.32

Table 2. **Cumulative ablation on MedicinalPlant.** Each row adds one component to the preceding setting; all rows use the same Agri-DINO backbone. Gain is the change from the preceding row.

Cumulative setting	Res.	Acc. (%)	Gain (pp)
(a) No LP, 224^2 , no TTA	224^2	35.19	—
(b) + high-resolution tokens	256^2	39.82	+4.63
(c) + LP warmup	256^2	52.41	+12.59
(d) + LLRD ($\gamma=0.75$)	256^2	57.88	+5.47
(e) + horizontal-flip TTA (full)	256^2	59.67	+1.79

over random initialization. Its score is **0.32 points** below the reported ImageNet-21k baseline (**59.99%**). Because the baseline rows use direct fine-tuning while Agri-DINO uses LP-FT, this comparison indicates competitive transfer under the reported protocols; it does not isolate pre-training from adaptation or establish equivalence across datasets and deployment conditions.

4.3. Ablation Study

Table 2 reports a cumulative ablation of the proposed fine-tuning protocol. All rows use the same Agri-DINO backbone; each row adds one component to the preceding configuration.

The cumulative ablation shows three practical effects. Increasing the input resolution from 224^2 to 256^2 adds **+4.63 points** (row b), consistent with the value of fine structure for botanical discrimination. Adding the LP warmup produces the largest incremental gain, **+12.59 points** (row c), which is consistent with early head–backbone mismatch being an important failure mode of direct adaptation. LLRD adds a further **+5.47 points** (row d), while horizontal-flip TTA adds **+1.79 points** at inference time and requires no additional training. Because the rows are cumulative, these increments should not be interpreted as independent effects.

4.4. Convergence Analysis

Table 3 traces validation accuracy and loss at selected checkpoints for the full Agri-DINO model. The two-stage transition at epoch 5 (LP $\rightarrow$ LLRD) is followed by a sharp rise in validation accuracy as backbone gradients begin propagating, followed by a gradual plateau.

Table 3. **Training dynamics of Agri-DINO on MedicinalPlant.** Epoch 5 marks the LP $\rightarrow$ LLRD transition. Accuracies are top-1; validation loss is cross-entropy. Values are shown at selected checkpoints.

Epoch	Train Acc (%)	Val Acc (%)	Val Loss
1	22.3	24.1	3.401
5	41.8	39.2	2.994
10	62.4	51.6	2.811
20	73.6	56.3	2.694
30	78.8	58.1	2.621
40	82.9	59.4	2.606
50	84.4	59.7	2.598

Table 4. **PlantSeg audit under fragmented vs. corrected taxonomy.** Reported results use the fragmented 300-class conversion; corrected 20-category results are pending relabeling. All models use ViT-Base/16 at 256×256 .

Method	Pre-train	Val Acc %	
		Frag. (300 cl.)	Corr. (20 cl.)
Scratch	—	2.42	—
ImageNet-21k	Sup.	2.78	—
Agri-DINO (Ours)	SSL	2.63	—

Corrected results pending relabeling.

Across the sampled checkpoints, validation loss decreases smoothly, with no divergence in the reported run. The train-validation accuracy gap at epoch 50 is approximately 25 points, which indicates substantial residual overfitting on this 1.3K-image dataset rather than its absence. The protocol therefore improves optimization stability, but it does not eliminate the need for stronger split control and regularization in future experiments.

4.5. Analysis: Micro-Class Fragmentation in PlantSeg

All three initialization methods achieve only $\approx 2.6\%$ accuracy on PlantSeg (Table 4). Rather than interpreting this number as evidence that the visual task is intrinsically intractable, we audited the conversion pipeline and found a systematic benchmark pathology.

The automated conversion pipeline parsed sample-level file identifiers (e.g., `banana-black-sigatoka-10-` and `banana-black-sigatoka-102-`) as distinct taxonomic classes, inflating the label space from the true **20** disease categories to **300 micro-classes**. With only a few hundred images, most classes then contain fewer than three training examples. The random-chance baseline is $100/300 \approx 0.33\%$; the observed 2.42–2.78% is only modestly informative given this fragmentation. We are correcting

the pipeline by stripping numeric suffixes before taxonomic assignment, but corrected-taxonomy results are not yet available. More broadly, any conversion from YOLO annotations to image-classification folders should separate instance identifiers from semantic labels explicitly.

5. Discussion

In-domain self-supervision. On the reported MedicinalPlant split, Agri-DINO (59.67%) is 0.32 points below supervised ImageNet-21k initialization (59.99%), despite using no downstream human labels during additional pre-training. This result suggests that unlabeled agricultural images can provide a competitive initialization when labeled botanical data are scarce. The comparison is protocol-dependent and should not be extrapolated to field robustness or to other taxonomies without external and group-disjoint evaluation.

Why LP-FT helps in this setting. The LP warmup contributes **+12.59 points** in our cumulative ablation, the largest single increment. A plausible explanation is that the randomly initialized head is initially mismatched to DINO’s view-invariant feature space; fitting the head first reduces that mismatch before backbone updates begin. This interpretation is consistent with the ablation, but it is not a direct measurement of feature drift or a universal property of all self-supervised backbones.

Limitations and future directions. Four limitations bound the current conclusions. First, evaluation is limited to image classification on one medicinal-plant split; it does not test lesion segmentation, oriented detection, counting, or cross-farm generalization. Second, the reported comparison uses one split and one run, so the 0.32-point difference to ImageNet-21k should not be treated as statistically meaningful without repeated trials. Third, all experiments use RGB images. Near-infrared and related spectral channels can expose early stress signals, motivating a lightweight adapter for multispectral input. Fourth, 256×256 remains modest for small lesions, while scaling toward 448×448 increases quadratic attention cost. Future work should combine repeated, leakage-safe evaluation with higher resolution, real multispectral data, and external-domain tests.

6. Conclusion

We presented Agri-DINO, a DINOv3-initialized Vision Transformer adapted to fine-grained plant recognition with additional agricultural self-supervision and a two-stage LP-FT protocol. On one stratified medicinal-plant split, the reported Agri-DINO run improves accuracy by **+10.32 points** over random initialization and is close to the ImageNet-21k baseline under different adaptation protocols. The cumulative ablation identifies the incremental gains from resolution, LP warmup, LLRD, and TTA, while the PlantSeg audit ex-

poses an annotation-conversion error that fragments semantic labels into sample-specific classes. These results motivate broader evaluation, not a claim of universal agricultural superiority. Repeated seeds, leakage-safe splits, corpus-overlap audits, corrected PlantSeg labels, and external-domain tests are needed to establish how far the observed gains generalize.

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